Innovative Academy
Future Redefined

C++ PROGRAMMING

Course Duration: 40 hrs

S. No	Content
1	**INTRODUCTION** 1. Introduction to C++ Programming 2. History and Overview of C++ 3. Importance of C++ in software development 4. Applications of C++ 5. Basic Structure of a C++ Program 6. Basic Input/Output. 7. Example Program
2	**Variable** • Naming rules • Variables declaration **Datatypes** • Primitive Datatype • Non-Primitive Datatypes **Constants** • What is constants • Creating constant variable… • Using const keyword • Using #Define Preprocessor
3	**Operators** 1. Arithmetic

	2. Assignment 3. Relational 4. Misc 5. Logical 6. Bitwise 7. Conditional
4	**Decision Making (Branching)** 1. If Statement 2. If – Else 3. Nested If 4. If else if Ladder 5. Switch case
5	**Flow Control** 1. For Loop 2. While Loop 3. Do – While Loop
6	**Jump Statement** 1. Break 2. Continue 3. goto
8	**Functions** • Advantage of Functions • Types of Functions • Namespace • Virtual function

	• Friend function
9	**Array** • Initialization of an array • Accessing array elements • Types of an array 1. **Single Dimensional Array** 2. **Multi-Dimensional Array** a) Two-Dimensional b) Three-Dimensional • Sample program using array
10	**String** • Introduction to Strings • Array of String • Standard String Library Functions • Sample Program
11	**Pointer** • Introduction to Pointers • Pointers attribute • Pointers as function parameter • Pointers and Array • Pointers to pointers
12	**Dynamic Memory** • Introduction to dynamic memory allocation • Stack memory allocation.

	• Heap memory allocation.
13	**Structure** • Introduction to Structure • Using struct keyword • Sample Program **Union** • Introduction to Union • Difference between structures and unions • Using union keyword • Sample Program
14	**Enum** • Introduction to Enum • Using enum keyword • Sample Program **TypeDef** • Introduction to TypeDef • Using typedef keyword • Sample Program
15	**Oops** 1. Class and Objects • Defining classes and objects • Creating classes and objects.

	2. Access Modifier • What is access modifier • Need of access modifier 3. Constructor • Defining and creating constructor • Types of constructor
16	**Inheritance** • Introduction to inheritance • **Types:** Single inheritance • Multiple inheritance • Multi-level inheritance • Hierarchical inheritance • Hybrid inheritance
17	**Polymorphism** • Introduction to polymorphism • Compile time polymorphism 1. Function Overloading 2. Operator Overloading • Runtime polymorphism **1.** Virtual function
18	**Abstraction** • Introduction to abstraction • Creating abstract class and function • Using abstract keyword **Encapsulation** • Introduction to Encapsulation • Using setter and getter function

	• Example program.
15	**Exception Handling** • Introduction to Exception handling • Defining the Exception • Using Try block to detect error • Using Catch block to handle error • Throw used to pass the error
16	**File I/O Handling** • Introduction to file handling • Creating a file in system • Using header files <iostream> & <fstream> • Ofstream to open file to write • write data using getline() function • ifstream to read only • Example Programs.
17	**Exercise & Practice**